

Sayan Ghosh

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EDUCATION

VIT Bhopal University

B.Tech, Computer Science and Engineering (Cybersecurity)

Bhopal, MP, India

Aug. 2023 – May 2027

CBSE Board (Class XII)

PCM+CS

April 2023

INTERNSHIPS

Software Developer Intern

June 2026 – August 2026

UnsaidTalks Education Pvt. Ltd.

Remote

- Engineered modular database access for the Job Board module using Prisma ORM and NestJS, reducing schema setup overhead and establishing reusable data access patterns across environments
- Refactored legacy backend modules into a NestJS architecture under senior guidance, verifying CRUD reliability and schema validation across staging environments
- Built React frontend components for sot.unsaidtalks.com using the TanStack Query and Router ecosystem

PROJECTS

SentinelX IDS – AI-Powered Intrusion Detection System | *Python, FastAPI, React, SQLAlchemy, WebSockets*

- Built a full-stack intrusion detection system with a FastAPI backend (60+ REST endpoints), React dashboard, and real-time WebSocket alerting, secured with JWT auth, role-based access control, and an audit trail
- Engineered a Sigma-style detection engine with 8 YAML rules mapped to MITRE ATT&CK, a sliding-window event correlator, and statistical anomaly detection (Z-score, IQR) across 5 log parsers
- Integrated 5 threat-intel feeds with GeoIP enrichment and risk scoring, a multi-provider LLM copilot for alert triage, and Slack/Discord/email notifications, with GitHub Actions CI running pytest

CipherMind – Security-Focused RAG Platform | *Python, FastAPI, ChromaDB, Groq (Llama 3.1), Hugging Face*

- Built and deployed on Render a RAG platform using FastAPI, ChromaDB vector storage, and Hugging Face embeddings to ground Llama 3.1 responses in retrieved document chunks
- Designed separate general and cybersecurity knowledge bases (OWASP Top 10, Burp log templates) with a security-auditing persona, tested on SQL injection, privilege-escalation payloads, and path traversal
- Added resilience with retry and backoff for embedding cold starts, API-key failover on rate limits, capped session memory, and environment-based secrets management

Home Lab Threat Intelligence & Detection Dashboard | *Python, Go, PostgreSQL, Docker, Scapy, scikit-learn*

- Engineered a real-time network monitoring system with decoupled microservices: Python/Scapy packet capture, a Go REST ingestion API with input validation and parameterized SQL, and PostgreSQL with INET types
- Implemented an unsupervised Isolation Forest pipeline over a 60-minute rolling window, using per-destination features (packet count, average size, total bytes) to flag anomalous endpoints such as potential C2 beaconing
- Built a SOC-style Streamlit dashboard with a 5-second refresh loop, containerized with Docker Compose

TECHNICAL SKILLS

Languages: Java, Python, Go, C++

Frameworks: Spring Boot, React, NestJS, FastAPI

Databases: PostgreSQL, MySQL, Redis, ChromaDB

Tools & Platforms: Docker, Git, GitHub Actions, Vercel, AWS

Security/AI: OWASP Top 10, MITRE ATT&CK, RAG, LLM APIs, scikit-learn, pandas, NumPy

CERTIFICATIONS & ACHIEVEMENTS

Supervised Machine Learning – Coursera / DeepLearning.AI

Google IT Support Professional Certificate – Google / Coursera

Oracle Cloud Infrastructure Certified Data Science Professional – Oracle

Solved 300+ data structures and algorithms problems across LeetCode and other coding platforms